INTERNET OF THINGS · CHART 2

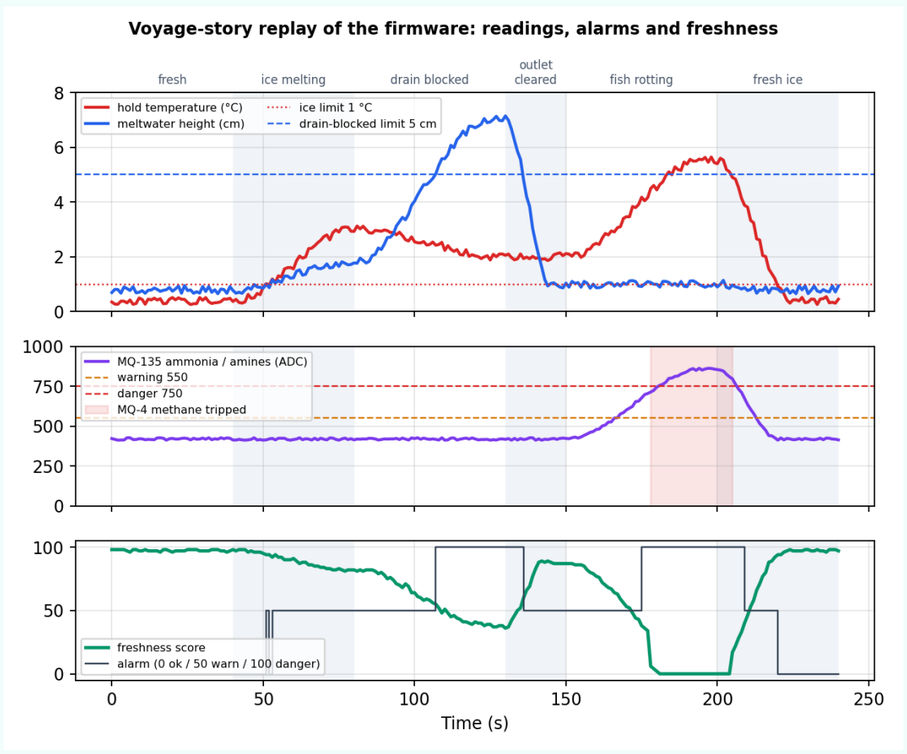
NodeMCU ESP8266

Fish Quality Monitor

Freshness score, box display codes, phone dashboard and a simulated voyage

Join WiFi Boat-Fish-Monitor → open 192.168.4.1 → see the hold live

1 Test: a simulated voyage



The firmware's own simulator: ice melts, the drain blocks, fish rot, then fresh ice.

2 Box display codes

Box shows	Meaning	Do this
ICE	hold too warm	add crushed ice
drAn	meltwater too high	clear the drain
nH3	ammonia high	check for rotting fish
CH4	methane	remove spoiled fish
E-dH, E-US, E-nH	sensor not answering	check the wire

3 Freshness score

Freshness starts at 100, minus	Pts
each °C above 0 °C	6
each cm of water above 2 cm	10
MQ-135 above clean air (per 7)	1
MQ-4 methane trip	25

80+ **FRESH** · 60+ **GOOD** · 40+ **CHECK** · below 40 **SPOILED**

4 What the program does

1. Read DHT11, HC-SR04, MQ-135, MQ-4
2. Sensor missing? -> E-code, NO SENSOR
3. Compare with the limits
4. Work out the freshness score
5. All fine -> slide through summary
6. Problem -> flash code + red LED
7. Log a row every minute

5 Wiring

Part	NodeMCU pin
TM1637 CLK / DIO	D5 / D6
DHT11	D4
HC-SR04 TRIG / ECHO	D7 / D8*
MQ-135 AO	A0
MQ-4 DO	D1
Red LED	D0

* ECHO through a 1 kΩ / 2 kΩ divider.

6 Logging & simulator

- CSV log in flash; download it from the phone.
- Laptop **simulator**: sliders beside the dashboard.
- Voyage story plays every alarm in 4 min.

7 Next steps

DS18B20 probe for exact ice temperature · GSM / LoRa alerts to the owner on shore · automatic drain pump